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(54) **INFRASTRUCTURE DISTRIBUTED DENIAL
OF SERVICE PROTECTION**

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H04L 12/721 (2013.01)

H04L 45/12 (2022.01)

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(2013.01); **H04L 63/029** (2013.01); **H04L**
63/10 (2013.01); **H04L 63/1408** (2013.01);
H04L 45/12 (2013.01); **H04L 63/1441**
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(56) **References Cited**

U.S. PATENT DOCUMENTS

9,363,102 B1 * 6/2016 Brandwine H04L 12/54

9,407,539 B1 * 8/2016 Dickinson H04L 45/306

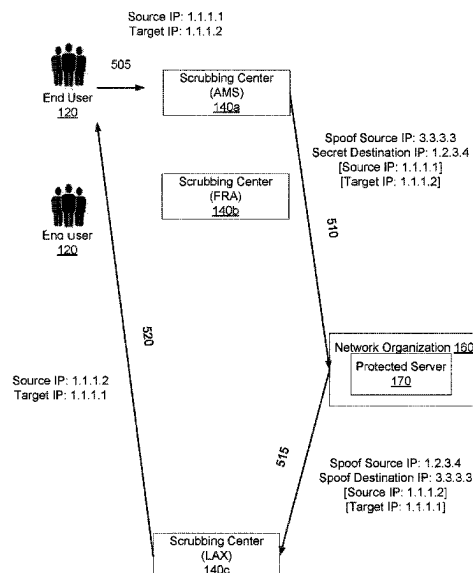
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Primary Examiner — Noura Zoubair

(57) **ABSTRACT**

A method of providing infrastructure protection for a server of a network organization, the method including announcing, as an internet protocol (IP) address associated with a server of a plurality of servers, a first anycast IP address, the first anycast IP address being one of a plurality of anycast IP addresses that each serve as an anycast address for a scrubbing center network. Each of the plurality of anycast IP addresses is allocated to a respective server of the plurality of servers by the scrubbing center network. The scrubbing center network may receive an incoming network packet intended for the server, the incoming network packet identified using the first anycast IP address. The scrubbing center network may determine whether the incoming network packet is legitimate and if so, the incoming network packet may be routed to the server using a generic routing encapsulation (GRE) tunnel.

14 Claims, 5 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 62/353,021, filed on Jun. 21, 2016.

References Cited

(56)

U.S. PATENT DOCUMENTS

2015/0156035 A1 * 6/2015 Foo H04L 12/4645
370/392
2015/0222528 A1 * 8/2015 Lotfi H04L 67/14
709/203
2016/0241590 A1 * 8/2016 Smith H04L 67/1001

* cited by examiner

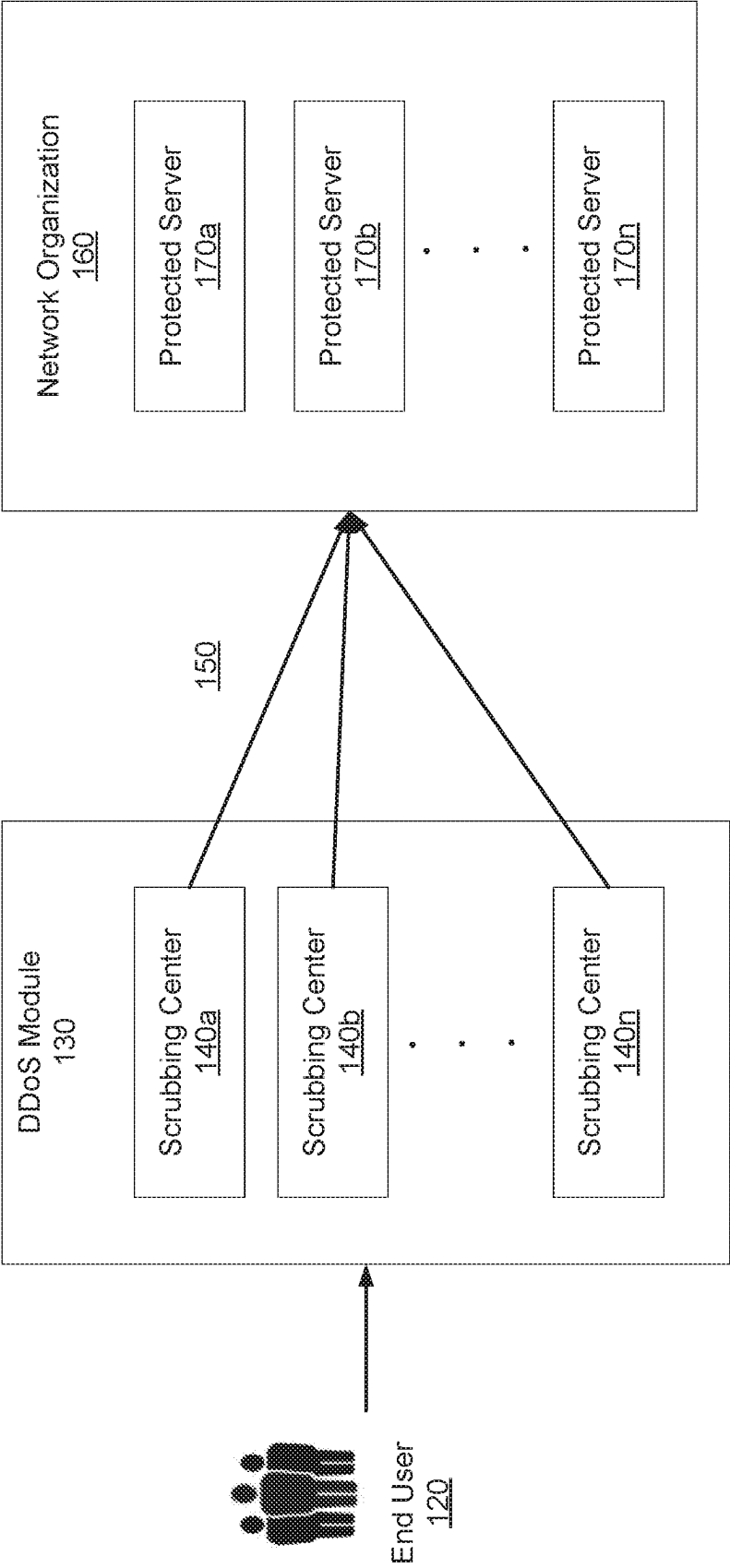
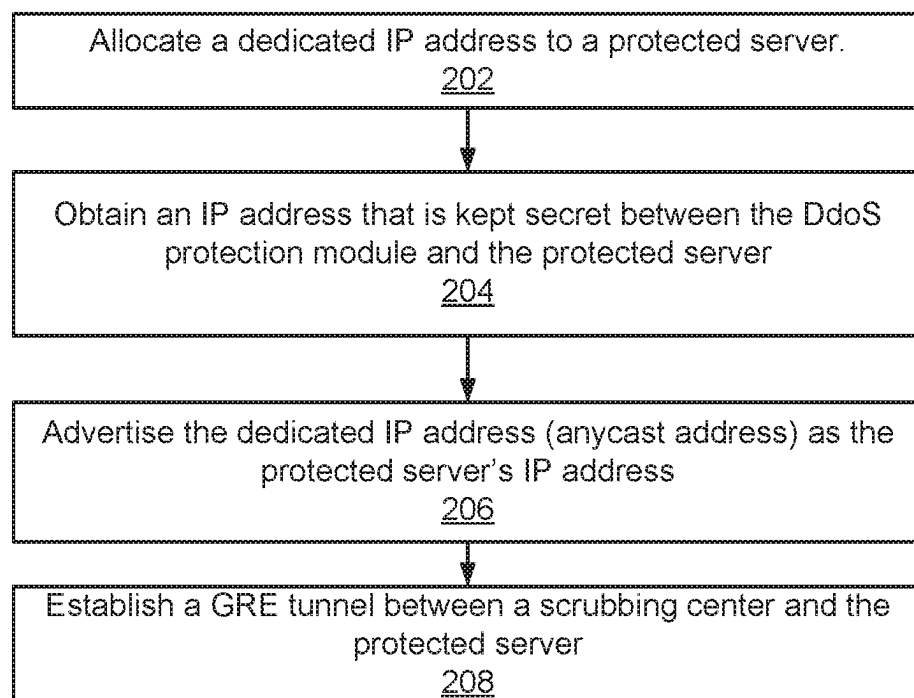


FIG. 1

200**FIG. 2**

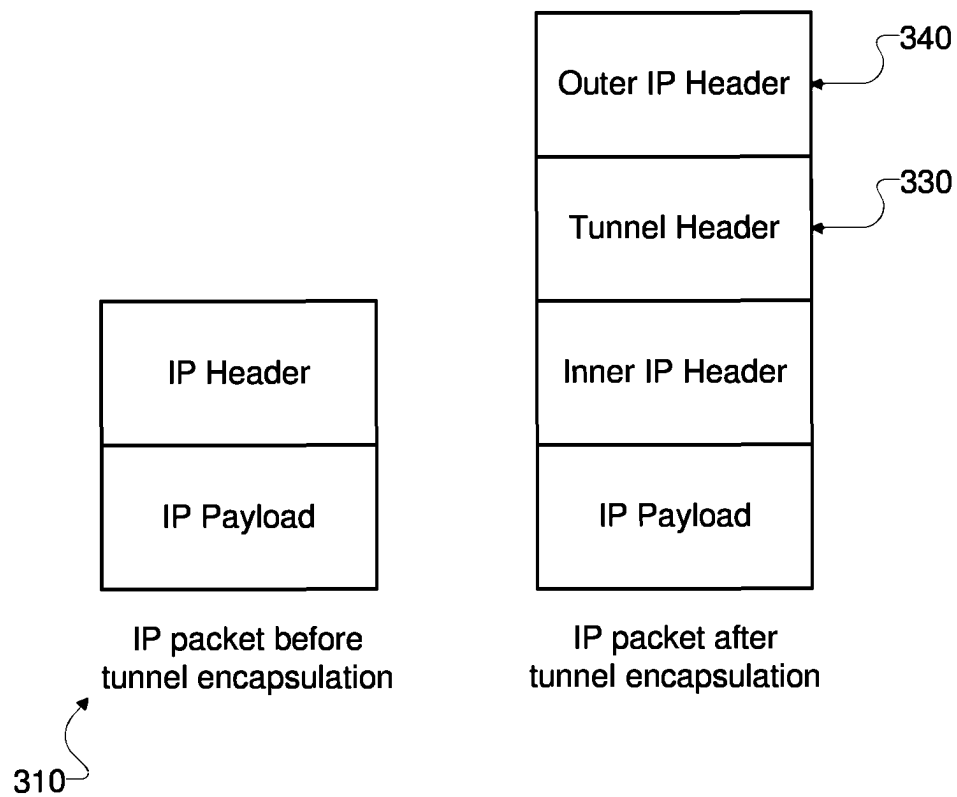


FIG. 3

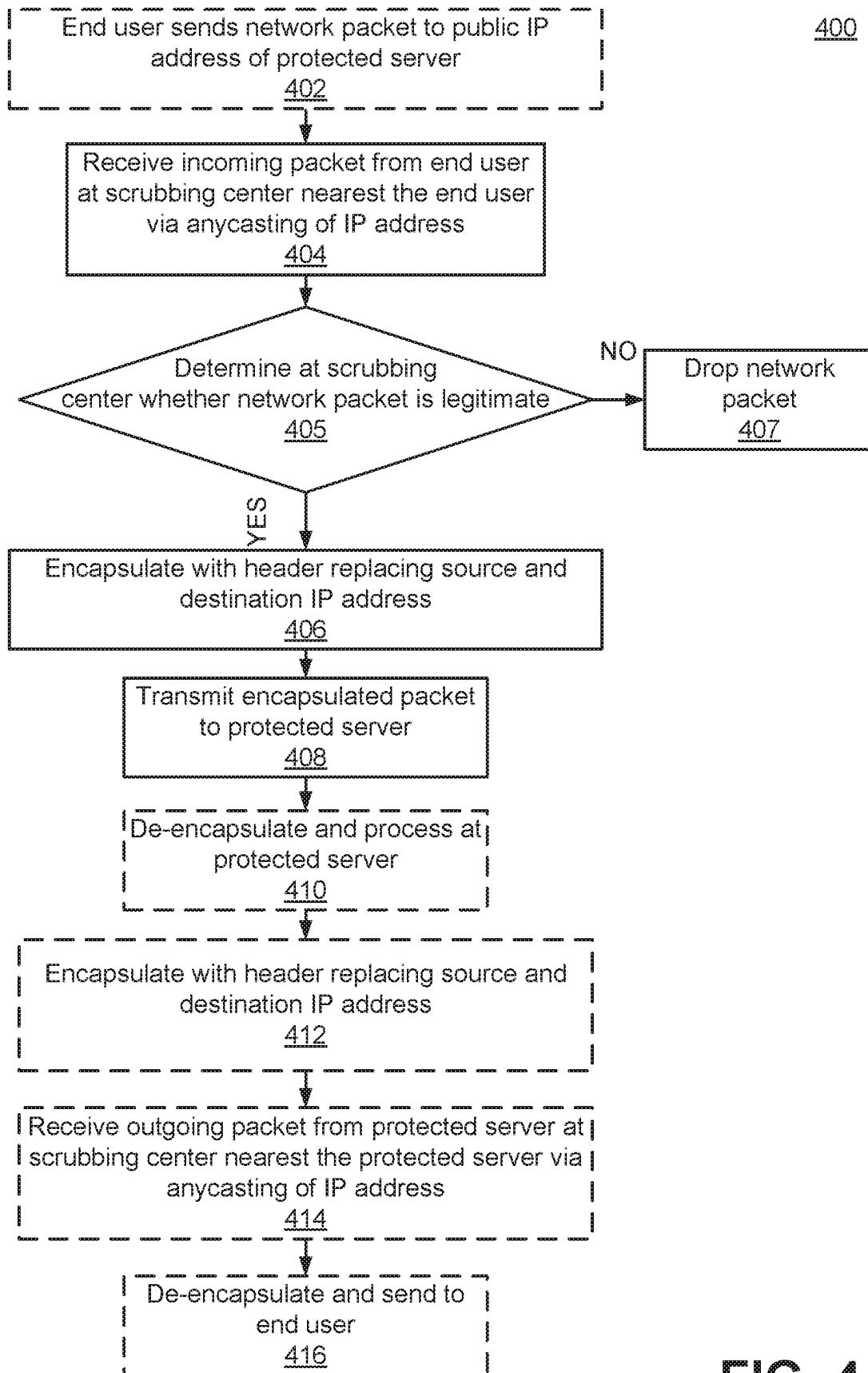
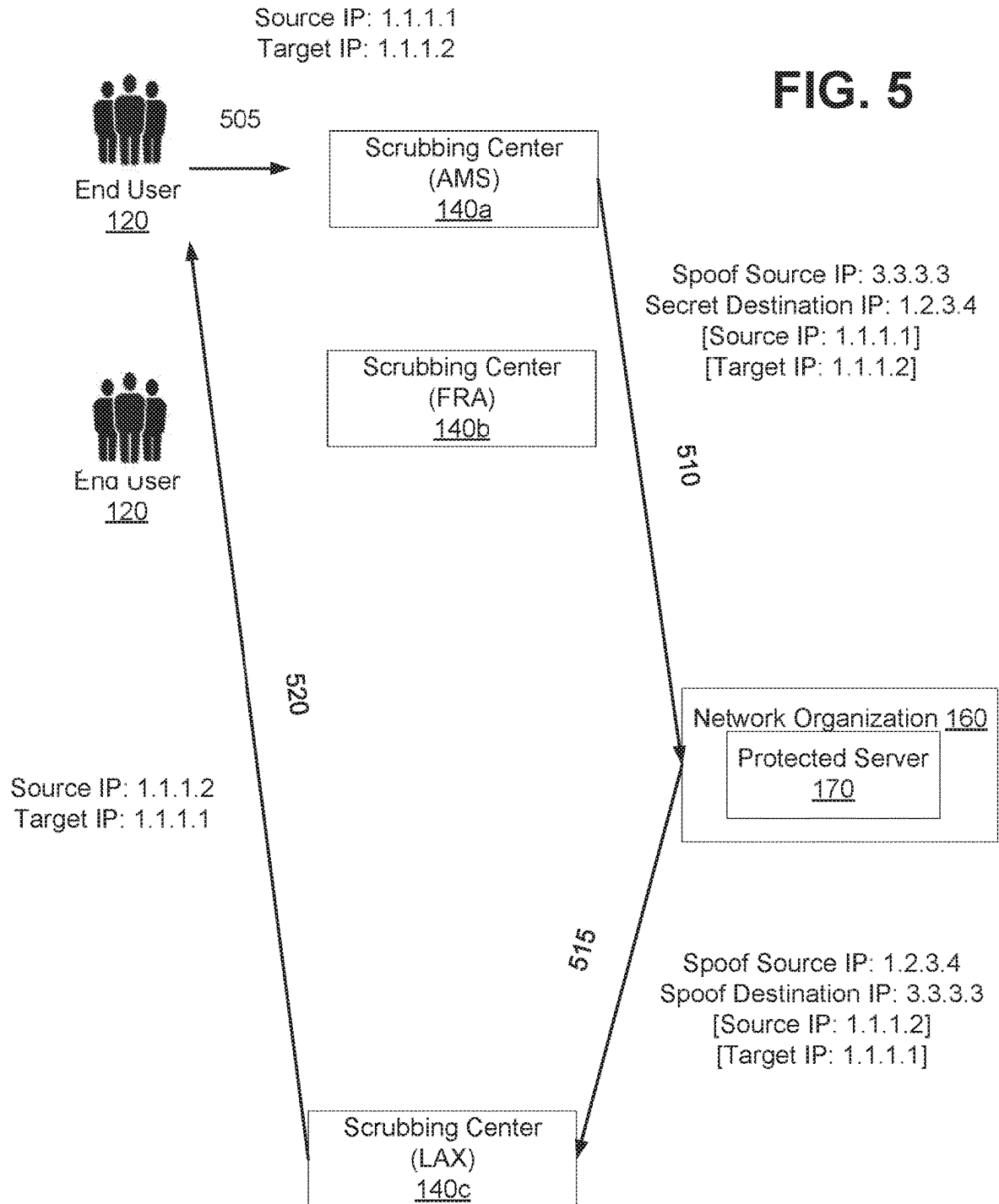


FIG. 4



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INFRASTRUCTURE DISTRIBUTED DENIAL OF SERVICE PROTECTION

RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 16/839,666, filed Apr. 3, 2020, which is a continuation of U.S. application Ser. No. 16/749,883, filed Jan. 22, 2020, which is a continuation of U.S. application Ser. No. 15/628,620, filed Jun. 20, 2017, now issued as U.S. Pat. No. 10,574,691 on Feb. 25, 2020, which claims the benefit of U.S. Provisional Application No. 62/353,021, filed on Jun. 21, 2016, all of which are incorporated herein by reference in their entirety for all purposes.

FIELD OF DISCLOSURE

This disclosure relates to security in a computing system, and in particular to Distributed Denial of Service (DDoS) protection for a network infrastructure.

BACKGROUND

Infrastructure protection services allow organizations to tunnel all ingress traffic (traffic from the Internet to the origin network) through a scrubbing centers network. One or more edge routers of an organization's network use the Border Gateway Protocol (BGP) to announce subnets and IP ranges to be advertised by the scrubbing centers, thus forcing all incoming traffic via the Internet routes to point at the scrubbing centers network instead of pointing at the organization's data center that forms the network infrastructure.

Infrastructure protection services use Generic Routing Encapsulation (GRE) tunneling to forward the incoming traffic to the organization's network after the traffic has been scrubbed from any DDoS attack. The incoming/outgoing Internet traffic for an organization can be routed through scrubbing centers in two ways. The first is by an organization's edge router announcing that the scrubbing centers' autonomous system (AS) is the owner of the organization's IP range. The second is done by the scrubbing centers edge routers announcing that they are the owner of the organization's IP range. Both of these methods lead to the customer traffic being routed through the scrubbing centers and scrubbed prior to reaching the organization's data center.

In order to use the infrastructure protection service, an organization must own a minimum number of IP addresses (for example, 256 IP addresses). Thus, an organization with IP addresses less than the minimum number required for traditional infrastructure protection services may not be able to avail the service, and thus may be prone to DDoS attacks. Additionally, existing systems require that each protected network manually establish a dedicated GRE tunnel with each of the scrubbing centers. This is very difficult for smaller organizations that lack the proper network administration skills.

Prior solutions have been sub-optimal, including: On-premises solutions, which are rapidly losing their advantage as attacks get larger, and bots get more human-like. It makes more sense to mount a community defense, in the cloud. However, cloud-based solutions are better equipped to handle the changing attack environment, but require that you protect an entire Class C network. ISP clean pipe solutions lack capacity to handle just average size attacks, especially when it comes to packet-based attacks. Security isn't always a core capability of an ISP and they often don't have the security expertise of a dedicated DDoS mitigation provider.

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Proxy-based solutions completely hide the client IP, which breaks many applications and completely bypasses the fire-wall. Hybrid solutions are transient, only kicking in when a sustained attack occurs.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an environment for a DDoS protection module, according to one embodiment.

FIG. 2 illustrates a flow of events when an organization's network onboards the infrastructure protection services via the DDoS protection module, according to one embodiment.

FIG. 3 illustrates an IP packet before adding a GRE header and after adding a GRE header, according to one embodiment.

FIG. 4 illustrates a flow of events that occur when an end user accesses an organization's network, according to one embodiment.

FIG. 5 illustrates an example of incoming and outgoing traffic from and to an end user and a protected server, according to one embodiment.

The figures depict various embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

FIG. 1 is a block diagram that illustrates an environment for a DDoS protection module 130, according to one embodiment. The environment includes one or more end users 120, a DDoS module 130 with one or more scrubbing centers 140 and a network organization 160 that includes one or more protected servers 170. The scrubbing centers 140 and network organization 160 may interface, e.g., via a virtual GRE tunnel 150. In alternative configurations, different and/or additional components may be included in the system environment.

The environment includes one or more end users 120. The end users 120 are one or more computing devices capable of receiving user input as well as transmitting and/or receiving data to and from an organization's network infrastructure via a network. In one embodiment the end user 120 may be an electronic device such as a smartphone, a tablet computer, a desktop computer, or a laptop computer. Alternatively, an electronic device may be a device having computer functionality that accesses a set of mobile applications. In another embodiment, the end user 120 may a potential malicious DDoS attacker, i.e. and individual trying to hack the organization's network infrastructure or an electronic device program that tries to hack the network infrastructure.

The network over which the end user 120 communicates may comprise any combination of local area and/or wide area networks, using both wired and/or wireless communication systems. In one embodiment, the network uses standard communications technologies and/or protocols. For example, the network includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, code division multiple access (CDMA), digital subscriber line (DSL), etc. Examples of networking protocols used for communicating via the network include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol (HTTP), simple mail transfer protocol (SMTP), and file transfer

protocol (FTP). Data exchanged over the network may be represented using any suitable format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network may be encrypted using any suitable technique or techniques.

The DDoS protection module 130 provides infrastructure protection service for a network infrastructure that has IP addresses as low as a single IP address. The DDoS protection module 130 may be a point-of-presence (POP) that includes scrubbing centers 140, routers, switches, routing servers and other such components. A scrubbing server's web protection is based on a network of secure reverse proxies deployed on a globally distributed CDN. Web traffic that is routed through the scrubbing centers 140 network is terminated by those proxies, allowing scrubbing centers 140 to inspect each and every request to the website and identify and block any malicious activity. Organizations using web protection update their domain DNS to point to a unique hostname (CNAME) provided by the scrubbing centers 140 (e.g., *mysite.incapdns.net*).

The technologies utilized for providing the infrastructure protection are enabled by one or more scrubbing centers 140. The DDoS module 130 may employ or multiple scrubbing centers 140a-n. The scrubbing centers 140 run innovative algorithms and software technology on standard off-the-shelf hardware. A scrubbing server appliance on the scrubbing center 140 performs all Layer 3 and Layer 4 DDoS scrubbing and then passes all legitimate traffic back to an organization's network infrastructure 160, e.g., using a routing server. The scrubbing server is a powerful network layer processing module that is capable of processing incoming data packets at a very high speed—much faster than the target machines within the organization's network 160 that run target applications. The scrubbing center 140 is considered “nearest” the end user or protected server if it is geographically closest, lowest cost, healthiest, with the least congested route, and/or any other distance measure. The technology embedded in the routing server has unique routing capabilities that enable GRE tunnel 150 and packet processing, as described herein.

The organization's network infrastructure 160 includes one or more protected servers 170. The protected server 170 encapsulates an outgoing network packet with header information including the anycast public IP address, and sends it to the scrubbing center nearest the protected server via the anycast public IP address.

The generic routing encapsulation tunnel information comprises use of a single, virtual generic routing encapsulation tunnel 150 for the plurality of scrubbing centers 140. Likewise, a single generic routing encapsulation tunnel 150 can serve plurality of protected servers in the network organization 160.

The scrubbing centers 140 ensure that any scrubbing server 140 can handle any ingress packet and any egress packet. Scrubbing centers 140 “spoof” the source IP address from which they send the packets via the virtual GRE tunnel in the ingress section of the packet, such that an anycast address for the scrubbing servers is used as the source of the traffic from the perspective of the protected server. In short, all the scrubbing centers use the same origin IP address when sending packets to the protected server 170. This is a unique technique that causes all the scrubbing centers 140 to appear to be the same to the protected server 170 of an organization's network infrastructure 160. The protected server 170 sends the egress packets to the anycasting scrubbing centers 140 and is oblivious to which scrubbing center

140 actually receives it. The spoofing mechanism provides transparency to the protected server 170. From the protected server 170 side, it appears like it is communicating with a single scrubbing center 140.

FIG. 2 illustrates a flow of events when an organization's network 160 onboards the infrastructure protection services via the DDoS protection module 140, according to one embodiment. The DDoS protection module 140 owns a range of IP addresses. For each protected server of an organization's network infrastructure 160, the DDoS protection module 140 allocates 202 a dedicated IP address from within that range. The protected server further obtains 204 an IP address from an Internet Service Provider (ISP) that is kept secret between the DDoS protection module 140 and the protected server. The protected server may be one of a plurality of protected servers in the network organization.

The protected server advertises 206 the dedicated IP address as its address to the world using domain name system (DNS). Thus an end user 120 that identifies the protected server by NAME would use the dedicated IP address for communicating with the protected server. The dedicated IP address is advertised, as an anycast address. An anycast address includes addressing routes datagrams to a single member of a group of potential receivers that are all identified by the same destination address. This is a one-to-nearest association. Thus anyone trying to access the protected network would end up going through the nearest scrubbing center. The scrubbing center 140 is considered “nearest” the end user or protected server if it is geographically closest, lowest cost, healthiest, with the least congested route, and/or any other distance measure.

Further in the onboarding process, a “virtual GRE tunnel” is established 208 between the protected server of the organization's network 160 and the scrubbing centers network within the DDoS protection module 140. The “virtual GRE tunnel” is achieved by the fact that the IP address of the GRE tunnel in scrubbing centers side is published as an anycast IP which means that all of scrubbing center would wide publish it as their own. This leads to the fact that the closest scrubbing center to the packet to/from a protected server would service it.

The GRE tunnel connected between scrubbing centers and the protected server is defined by the following three IP addresses. First, a scrubbing center public IP address. This is a single dedicated IP address, anycasted globally, on the scrubbing centers network to which the GRE tunnel is connected. The end user 120 is provided this address as the public IP address of the protected server 170 it is trying to reach. In the example of FIG. 5, this is IP address 1.1.1.2.

Second, the protected server's routable, secret IP Address. This is the protected network's actual IP address, which is kept secret between the scrubbing center 140 and the protected network 170. In the example of FIG. 5, this is IP address 1.2.3.4. Third, a scrubbing center protect IP address is used as a spoofed source address from the protected server's perspective (in the example of FIG. 5, 3.3.3.3). This is a single dedicated IP address, anycasted globally, on the scrubbing centers network which serves as the new protected origin IP address.

After the packet is received by the scrubbing center 140 from an end user 120, it will be scanned and filtered and then a GRE header will be added, routing it to the protected server 170. The protected server 170 will receive the packet, decapsulate its GRE header, and process it. When sending data back the end user 120, a GRE header routing the traffic back to the scrubbing center 140 will be added by the protected server 170. The routing server in the scrubbing

center **140** which will receive the packet will decapsulate its GRE header, and route the traffic back to the end user **120**. This process is described in greater detail in conjunction with FIG. **4**.

Scrubbing Center Anycasting

The scrubbing centers **140** perform two types of anycasting to ensure redundancy and packet loss prevention. Anycasting an IP address includes associating addressing routes datagrams to a single member of a group of potential receivers that are all identified by the same destination address. This is a one-to-nearest association. A scrubbing center **140** is considered “nearest” the end user or protected server if it is geographically closest, lowest cost, healthiest, with the least congested route, and/or any other distance measure.

The first type refers to anycasting a public IP address towards end users **120**. The scrubbing centers **140** advertise the public IP address used to represent the protected server **170** from all scrubbing centers **140** simultaneously using anycasting. Traffic from users throughout the world is sent to scrubbing centers **140** that are closest to them, ensuring minimum latency. Moreover, if a scrubbing center **140** goes down it will stop advertising these addresses so traffic is sent to the next closest one, ensuring almost unlimited redundancy across the entire network.

The second type refers to anycasting the scrubbing center’s **140** public IP address towards the protected server’s **170** origin. The scrubbing centers **140** advertise the public IP address of the virtual GRE tunnel from all scrubbing centers **140** simultaneously using anycasting. This public IP address may be the same or different from the one anycasted to end users. Traffic from the protected server origin **170** is sent to the scrubbing center **140** that is closest to it ensuring minimum latency. Moreover, if this scrubbing center **140** goes down it will stop advertising these addresses so traffic is sent to the next closest one, ensuring almost unlimited redundancy across the entire network.

FIG. **3** illustrates an IP packet before adding a GRE header and after adding a GRE header, according to one embodiment. Before the GRE tunnel encapsulation, the IP packet **310** includes an IP header and an IP payload. The GRE tunnel is established to route the packet to the protected server and to protect the packet from DDoS attacks when it is routed. A tunnel header **330** and an outer IP header **340** are added to the IP packet to encapsulate the IP packet. The outer IP header includes a source IP address which is an anycast address of the scrubbing server and a destination IP address which is the secret IP address established between the protected server and the DDoS protection module **140**. The secret IP address does not appear on record as belonging to the organization’s network.

Using the examples from FIGS. **1** and **5**, if the end user **120** source IP address is 1.1.1.1, and its original destination address is 1.1.1.2, this information would be included in the original IP packet before encapsulation (**310**). Then, when the scrubbing center **140** sends the packet to the protected server **170**, it adds (**330**, **340**) a spoofed source (3.3.3.3, the scrubbing center anycast address from the protected server’s perspective) and destination address (1.2.3.4, the true/private IP address of the protected server **170**). Likewise, on the outbound packet from the protected server **170**, the encapsulation headers (**330**, **340**) would show the source IP address as the protected server (1.2.3.4; private, known to the scrubbing center) and destination IP address as the spoofed scrubbing center anycast address (3.3.3.3). The

encapsulation would be removed at the scrubbing center **140** to ensure that the packet gets back to its original sourced, the end user (1.1.1.1).

FIG. **4** illustrates a flow of events **400** that occur when an end user **120** accesses an organization’s network **160**, according to one embodiment. In this example, the organization has performed infrastructure onboarding, e.g., as described in conjunction with FIG. **2**. As an initial step, an end user **120** sends **402** a network packet to a destination address that is the public IP address associated with a protected server **170**. For example, the end user **120** may look up the IP address of the protected network using a name service. No public records exist that associate the protected network with the private IP address used between the scrubbing server **140** and the protected network **160**. The incoming network packet is received **404** from the end user **120** at the scrubbing center **140** nearest the end user **120**, despite the incoming network packet being intended for a protected server, e.g., **170**, of a network organization, e.g., **160**, identified using the public IP address, as the public IP address serves as an anycast address for a plurality of scrubbing centers **140** in a distributed network of scrubbing servers, the plurality of scrubbing servers **140** including the scrubbing center **140** nearest the end user **120**. The incoming network packet includes the public IP address as the source of the incoming packet such that the protected server **170** does not know which scrubbing center **140** sent the incoming network packet. The scrubbing center “nearest” the end user or protected server can be geographically closest, lowest cost, healthiest, with the least congested route, or any other distance measure.

The scrubbing center **140** receives and processes the incoming network packet traffic to filter out illegitimate traffic (for example, a DDoS attack type of traffic). The scrubbing center **140** nearest the end user **120** determines **405** whether the incoming network packet is legitimate. Traffic that is considered illegitimate or malicious is dropped **407** by the scrubbing center **140** and thus never reaches the protected network **160**.

Responsive to the scrubbing center **140** determining that the incoming network packet is legitimate, it encapsulates **406** the incoming network packet with header information including a destination address including a private IP address for the protected server **170**. This private IP address unknown to the end user **120**, and is kept secret between the scrubbing center **140** and the protected server **170**. For example, the encapsulation used is GRE according to one embodiment.

Then the encapsulated incoming network packet is transmitted **408** to the protected server **170** at the private IP address based on the header information, e.g., via a GRE tunnel.

The incoming packet is received at the organization’s network infrastructure **160** according to one embodiment. The original network packet information is decapsulated **410**, extracted, processed, and routed to the appropriate protected server **170** in the protected network **160**. The packet is routed as if it was sent by the end user **120** to the organization’s network infrastructure **160** directly. This process maintains the actual client IP information and allows the organization **160** to detect its actual source of the traffic.

With respect to outgoing network traffic, the organization’s network infrastructure **160** handles egress packets destined for the end user **120** (they can be detected by their source which is maintained throughout this process) as follows. The same encapsulation techniques, e.g., a GRE tunnel, is configured on one of the organization’s network

infrastructure **160** interfaces so that packets being sent back to the end users **120** are encapsulated with a GRE header in order to go through the GRE tunnel back to the routing servers. These packages are destined for the scrubbing centers **140** anycast public IP address that is advertised by the routing servers. Specifically, an outgoing network packet is encapsulated **412** by the protected server **170** or network organization **160** with header information including the anycast public IP address of the scrubbing servers **140**. The encapsulated packet is routed through the Internet to the scrubbing center **140** nearest the protected server **170** sending it, since the IP address of the scrubbing center **140** interface is advertised as anycast. For example, in some circumstances the scrubbing center **140** nearest the end user **120** and the scrubbing center **140** nearest the protected server **170** are different scrubbing centers **140**.

The scrubbing center **140** receives **414** the outgoing network packet (e.g., including any GRE tunnel information). The scrubbing center **140** nearest the protected server **170** then decapsulates **416** the outgoing network packet such that the destination address is the end user **120**, and transmits the packet to the end user **120**.

FIG. 5 illustrates an example of incoming and outgoing traffic from/to an end user **120** and a protected server **170**, via scrubbing servers **140** of a DDoS protection module, according to one embodiment.

In the illustrated example, incoming traffic **505** from an end user **120** with a source (end user) IP address (1.1.1.1) attempts to contact protected server **170** an organization's network infrastructure **160** in Los Angeles, via a destination address that is the advertised public IP address for the server **170** (1.1.1.2). Three scrubbing centers **140a-c** are shown, which can receive and process the incoming traffic from the end user **120**, in Amsterdam **140a**, Frankfurt **140b**, and Los Angeles **140c**. The advertised public IP address (1.1.1.2) serves as an anycast address for the scrubbing centers **140** rather than the true IP address of the protected server **170**, which has kept secret between the scrubbing centers **140** and the protected server **170**. Thus, traffic directed to the public IP address (1.1.1.2) for protected server **170** instead is directed to the scrubbing center **140** nearest the end user **120**.

If the end user **120** is in Amsterdam, for example, the nearest scrubbing center **140** is in Amsterdam, **140a**, so it receives the traffic **505** from end user **120** intended for protected server **170** via the anycast IP address. Header information is added, e.g., by use of a GRE tunnel **150** between the scrubbing centers **140** and the protected server **170**. The header information replaces the destination address (the public IP address, 1.1.1.2) with the true, private IP address of the protected server (1.2.3.4), so that the traffic is delivered **510** to the protected server **170**, once it has been verified by the scrubbing center **140a** as legitimate. It also replaces the source address of the end user **120** (1.1.1.1) with a spoofed source IP address (3.3.3.3) that is an anycast address for the scrubbing servers **140** from the protected server **170** perspective.

In the illustrated example, outgoing traffic from the protected server **170** of the organization's network **160** to the end user **120** also is routed via the scrubbing centers **140** of the DDoS protection module, according to one embodiment. Continuing the example from above, when the Los Angeles protected server **170** receives the network packet, it will decapsulate its header and process it. When it sends back an outgoing response **515** intended for the end user's **120** IP address (1.1.1.1), it goes through the spoofed anycast IP address (3.3.3.3), and thus the outgoing traffic will be routed

via the scrubbing center **140** nearest the protected server **170**, scrubbing center **140c**. The protected server **170** uses the true source IP address (the private, protected server address 1.2.3.4) as the source IP address, since it is known to the scrubbing center **140**. The outgoing traffic eventually reaches **520** the end user **120** in Amsterdam, but this time via the Los Angeles scrubbing center **140c**. At the scrubbing center **140**, the header is removed so that the original source IP address (the end user address, 1.1.1.1) is revealed and the packet is correctly routed back.

The GRE tunnel can serve more than one protected IP address in some embodiments, for example in case the customer router is serving two different IP addresses, which are protected by two unique private IP addresses, a single GRE tunnel for these two different IP addresses can be used.

Additional Considerations

The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

Any of the devices or systems described herein can be implemented by one or more computing devices. A computing device can include a processor, a memory, a storage device, an I/O interface, and a communication interface, which may be communicatively coupled by way of communication infrastructure. Additional or alternative components may be used in other embodiments. In particular embodiments, a processor includes hardware for executing computer program instructions by retrieving the instructions from an internal register, an internal cache, or other memory or storage device, and decoding and executing them. The memory can be used for storing data or instructions for execution by the processor. The memory can be any suitable storage mechanism, such as RAM, ROM, flash memory, solid state memory, and the like. The storage device can store data or computer instructions, and can include a hard disk drive, flash memory, an optical disc, or any other suitable storage device. The I/O interface allows a user to interact with the computing device, and can include a mouse, keypad, keyboard, touch screen interface, and the like. The communication interface can include hardware, software, or a combination of both, and can provide one or more interfaces for communication with other devices or entities.

Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.

Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program

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code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

Embodiments of the invention may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

The invention claimed is:

1. A method comprising:

announcing, as an internet protocol (IP) address associated with a server of a plurality of servers, a first anycast IP address to represent the server, the first anycast IP address being one of a plurality of anycast IP addresses, the first anycast IP address as an anycast address for a plurality of scrubbing centers for a scrubbing center network, the first anycast IP address allocated specifically to the server, wherein each of the plurality of anycast IP addresses is allocated to a respective server of the plurality of servers by the scrubbing center network, and wherein an actual IP address of the server is maintained confidential between the scrubbing center network and the server; receiving, at the scrubbing center network, an incoming network packet intended for the server, the incoming network packet identified using the first anycast IP address; determining, by the scrubbing center network, whether the incoming network packet is legitimate; in response to determining that the incoming network packet is legitimate, routing, by a processor, the incoming network packet to the server at the actual IP address of the server using a generic routing encapsulation (GRE) tunnel; anycasting a second anycast IP address towards the server from each scrubbing center of the scrubbing center network simultaneously; receiving, at a scrubbing center of the plurality of scrubbing centers, an encapsulated outgoing network packet from the server, wherein the encapsulated outgoing network packet is an outgoing network packet which

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has been encapsulated by the server with a header information comprising the second anycast IP address as a destination address;

decapsulating the outgoing network packet to generate a decapsulated outgoing network packet by removing the encapsulation added by the server that includes the header information, wherein the decapsulating is performed by a scrubbing center of the scrubbing center network that is nearest to the server; and

transmitting the decapsulated outgoing network packet to an end user,

wherein the decapsulating is performed by a scrubbing center of the plurality of scrubbing centers that is nearest to the server.

2. The method of claim 1, wherein the GRE tunnel is defined by the first anycast IP address, the actual IP address of the server, and the second anycast IP address associated with the incoming network packet by the scrubbing center network.

3. The method of claim 2, wherein routing the incoming network packet to the server comprises:

modifying a header of the incoming network packet to indicate the second anycast IP address as a source IP address of the incoming network packet, the second anycast IP address being one of a second plurality of anycast IP addresses that each serve as an anycast address for the scrubbing center network.

4. The method of claim 1, wherein a mapping of the first anycast IP address to the actual IP address is maintained between the scrubbing center network and the server.

5. The method of claim 1, wherein the first anycast IP address is announced by the scrubbing center of the plurality of scrubbing centers nearest to an end user.

6. A system comprising

a memory; and

a processor operatively coupled to the memory, the processor to:

announce, as an internet protocol (IP) address associated with a server of a plurality of servers, a first anycast IP address to represent the server, the first anycast IP address being one of a plurality of anycast IP addresses, the first anycast address serves as an anycast address for a plurality of scrubbing centers for a scrubbing center network, the first anycast IP address allocated specifically to the server, wherein each of the plurality of anycast IP addresses is allocated to a respective server of the plurality of servers by the scrubbing center network, and wherein an actual IP address of the server is maintained confidential between the scrubbing center network and the server;

receive, at the scrubbing center network, an incoming network packet intended for the server, the incoming network packet identified using the first anycast IP address;

determine, by the scrubbing center network, whether the incoming network packet is legitimate;

in response to determining that the incoming network packet is legitimate, route the incoming network packet to the server at the actual IP address of the server using a generic routing encapsulation (GRE) tunnel;

anycast a second anycast IP address towards the server from each scrubbing center of the scrubbing center network simultaneously;

receive, at a scrubbing center of the plurality of scrubbing centers, an encapsulated outgoing network

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packet from the server, wherein the encapsulated outgoing network packet is an outgoing network packet which has been encapsulated by the server with header information comprising the second anycast IP address as a destination address;

decapsulate the outgoing network packet to generate a decapsulated outgoing network packet by removing the encapsulation added by the server that includes the header information, wherein the decapsulating is performed by a scrubbing center of the scrubbing center network that is nearest to the server; and

transmit the decapsulated outgoing network packet to an end user,

wherein the decapsulating is performed by a scrubbing center of the plurality of scrubbing centers that is nearest to the server.

7. The system of claim 6, wherein the GRE tunnel is defined by the first anycast IP address, the actual IP address of the server, and the second anycast IP address associated with the incoming network packet by the scrubbing center network.

8. The system of claim 7, wherein to route the incoming network packet to the server, the processor is to:

modify a header of the incoming network packet to indicate the second anycast IP address as a source IP address of the incoming network packet, the second anycast IP address being one of a second plurality of anycast IP addresses that each serve as an anycast address for the scrubbing center network.

9. The system of claim 6, wherein a mapping of the first anycast IP address to the actual IP address is maintained between the scrubbing center network and the server.

10. The system of claim 6, wherein the first anycast IP address is announced by a scrubbing center of the plurality of scrubbing centers nearest to an end user.

11. A non-transitory computer-readable medium having instructions stored thereon which, when executed by a processor, cause the processor to:

announce, as an internet protocol (IP) address associated with a server of a plurality of servers, a first anycast IP address to represent the server, the first anycast IP address being one of a plurality of anycast IP addresses, the first anycast address serves as an anycast address for a plurality of scrubbing centers for a scrubbing center network, the first anycast IP address allocated specifically to the server, wherein each of the plurality of anycast IP addresses is allocated to a respective server of the plurality of servers by the scrubbing center network, and wherein an actual IP address of the server is maintained confidential between the scrubbing center network and the server;

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receive, at the scrubbing center network, an incoming network packet intended for the server, the incoming network packet identified using the first anycast IP address;

determine, by the scrubbing center network, whether the incoming network packet is legitimate;

in response to determining that the incoming network packet is legitimate, route the incoming network packet to the server at the actual IP address of the server using a generic routing encapsulation (GRE) tunnel;

anycast a second anycast IP address towards the server from each scrubbing center of the scrubbing center network simultaneously;

receive, at a scrubbing center of the plurality of scrubbing centers, an encapsulated outgoing network packet from the server, wherein the encapsulated outgoing network packet is an outgoing network packet which has been encapsulated by the server with header information comprising the second anycast IP address as a destination address;

decapsulate the outgoing network packet to generate a decapsulated outgoing network packet by removing the encapsulation added by the server that includes the header information, wherein the decapsulating is performed by a scrubbing center of the scrubbing center network that is nearest to the server; and

transmit the decapsulated outgoing network packet to an end user,

wherein the decapsulating is performed by a scrubbing center of the plurality of scrubbing centers that is nearest to the server.

12. The non-transitory computer-readable medium of claim 11, wherein the GRE tunnel is defined by the first anycast IP address, the actual IP address of the server, and the second anycast IP address associated with the incoming network packet by the scrubbing center network.

13. The non-transitory computer-readable medium of claim 12, wherein to route the incoming network packet to the server, the processor is to:

modify a header of the incoming network packet to indicate the second anycast IP address as a source IP address of the incoming network packet, the second anycast IP address being one of a second plurality of anycast IP addresses that each serve as an anycast address for the scrubbing center network.

14. The non-transitory computer-readable medium of claim 11, wherein a mapping of the first anycast IP address to the actual IP address is maintained between the scrubbing center network and the server.

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